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Cross-scale Estimation of Aboveground Biomass in Arid Grasslands During the Peak Growing Season Using Ground Quadrats, UAV Multispectral Imagery, and Satellite Remote Sensing

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***Abstract*—Accurate estimation of grassland aboveground biomass (AGB) is essential for revealing carbon cycling mechanisms, assessing grassland degradation, and guiding ecological management. However, the complex background formed by the interspersion of vegetation and bare soil in arid areas, together with the spatial-scale mismatch between ground observations and satellite pixels, poses great challenges to accurate AGB estimation. Unmanned aerial vehicle (UAV) data with extremely high spatial resolution can effectively bridge the spatial scale gap between satellite pixels and ground measurements through upscaling, providing new opportunities for cross-scale AGB estimation in arid grasslands during the peak growing season. Based on ground plots, UAV, and satellite remote sensing data, this study integrated vegetation classification and multi-model comparison methods to construct a ground–UAV–satellite cross-scale inversion framework for estimating grassland AGB during the peak growing season in arid areas. The SHAP method was then used to interpret the contributions of key variables, and a spatial distribution map of AGB during the peak growing season was finally generated for the study area. The results showed that, at the UAV scale, the YOLOv8n-Seg model achieved the highest vegetation classification accuracy among the three grassland types (F1 = 0.761–0.800). After vegetation classification, AGB inversion accuracy was significantly improved, with the combination of PCA and ResNet achieving the best performance (R^2 = 0.847, RMSE = 149.24 kg/ha). At the satellite scale, the ground–UAV–satellite collaborative framework achieved better predictive accuracy for AGB estimation (R^2 = 0.718, RMSE = 202.15 kg/ha) than the commonly used ground–satellite framework. Overall, the proposed cross-scale collaborative estimation framework improved AGB inversion accuracy during the peak growing season in arid areas through vegetation classification, scale transformation, and variable-selection-based modeling, thereby providing a scientific basis for grassland AGB monitoring.**

***Index Terms*—Grassland, Aboveground biomass (AGB), Vegetation classification, Multiscale remote sensing, Deep learning.**

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I. INTRODUCTION

GRASSLANDS are an important component of terrestrial ecosystems [1] . They play a key role in maintaining biodiversity, regulating climate, providing food resources, and conserving water, while also delivering important ecosystem services such as carbon sequestration, biodiversity conservation, and the maintenance of ecological stability [2], [3]. The realization of these functions largely depends on the productivity level and spatial distribution of grassland vegetation. Aboveground biomass (AGB), as a key indicator for measuring grassland productivity, assessing ecosystem health, and characterizing environmental conditions, is a core input parameter in grassland ecological modeling and carbon cycle research [4]. Therefore, rapid and accurate estimation of grassland AGB is of great significance for improving our understanding of grassland carbon cycling processes, optimizing grassland resource management, and guiding ecological conservation planning.

Arid grassland ecosystems often exhibit a spatial pattern in which vegetation and bare soil are interspersed. This spatial heterogeneity makes vegetation information extraction susceptible to interference from the bare-soil background, thereby affecting the accuracy of aboveground biomass estimation [5]. Therefore, effectively distinguishing vegetation from non-vegetation areas and reducing background interference are critical for improving the accuracy of grassland AGB estimation [6]. In recent years, several studies have estimated grassland AGB using multi-source remote sensing data and machine learning methods, and have analyzed model performance and spatiotemporal variations [7], [8]. However, these studies have mainly improved AGB inversion accuracy from the perspectives of variable selection and model optimization, while insufficient attention has been paid to the mixed-pixel problem caused by sparse vegetation in arid areas, which may increase the uncertainty of grassland

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AGB estimation. Currently, the main methods for vegetation identification and extraction include feature-thresholding methods, machine learning, and deep learning [9]. Among them, vegetation-index-based thresholding methods are easy to implement and have been applied to the binary classification of vegetation and non-vegetation areas [10]. However, they have limitations in fine-scale classification because they cannot fully integrate multiple types of features. In contrast, machine learning and deep learning methods can automatically learn spectral and spatial features from images, reduce dependence on empirical rules, and show higher classification performance under complex land-surface backgrounds [11], [12]. Nevertheless, existing studies have still mainly focused on vegetation classification itself, while the extraction of shape and geometric features of classified vegetation remains insufficient.

In recent years, remote sensing-based grassland AGB estimation has increasingly focused on improving inversion accuracy through the optimization of model algorithms and variable selection. Compared with statistical models, machine learning (ML) and deep learning (DL) models have shown superior capability in AGB estimation. Among them, the random forest (RF) model has been successfully applied to biomass estimation in ecosystems such as wetlands, tundra, and alpine grasslands [13], [14], [15], while XGBoost has demonstrated strong performance in handling high-dimensional remote sensing features [16]. These models can integrate multi-source remote sensing data and adaptively address the complexity of grassland ecosystems, thereby improving model accuracy and generalization ability [17]. As an important branch of ML, deep learning (DL) has developed rapidly in recent years. Among various DL methods, ResNet, or residual network, introduces residual connection structures to automatically learn the complex nonlinear relationships between biomass and remote sensing inputs, and has achieved higher prediction accuracy than traditional methods [18]. Nevertheless, remote sensing-based biomass estimation still faces a key challenge, namely the substantial scale gap between ground sample data and satellite data. In field surveys, the mean value of several AGB quadrats is usually used to represent a sample [19], whereas the scale of these sample data is typically much smaller than the pixel scale of satellite data. This scale difference makes it difficult for satellite pixels to effectively capture the spatial heterogeneity of ground vegetation, thereby affecting the accuracy and reliability of AGB estimation.

The rapid development of unmanned aerial vehicle (UAV) technology has provided new opportunities for grassland AGB estimation. As a remote sensing tool with efficient and rapid data acquisition capability and strong operational capability in complex terrain, UAVs can carry multiple sensors and acquire centimeter-level resolution images, thereby finely characterizing grassland vegetation structure, patch distribution, and spectral features. However, due to limitations in flight endurance and coverage area, UAVs can only obtain high-precision data over relatively small study areas [20]. By contrast, satellite remote sensing provides large-area, continuous coverage and regional mapping capability, but its relatively coarse spatial resolution makes it difficult to fully capture vegetation heterogeneity at the ground-plot scale. Recent studies have shown that integrating satellite, UAV, and ground observations can improve vegetation health monitoring capability [21]. In addition, UAV and satellite data have complementary advantages in terms of spatial resolution and coverage area, providing support for multi-scale vegetation monitoring [22]. For vegetation biomass estimation, combining UAV observations with satellite imagery can enable regional-scale upscaling while preserving detailed surface information, thereby improving the accuracy of AGB estimation [23]. However, although previous studies have used UAVs to upscale ground observations to the satellite scale, studies that combine deep learning-based vegetation classification with ground–UAV–satellite cross-scale AGB estimation remain relatively limited, particularly for arid grasslands characterized by sparse vegetation, complex bare-soil backgrounds, and strong spatial heterogeneity [24].

Therefore, this study proposed a ground–UAV–satellite cross-scale collaborative estimation framework based on deep learning-based vegetation classification (Fig. 1). In this framework, UAV observations served as an intermediate scale to upscale ground-plot information to the satellite-pixel scale, thereby improving the regional inversion accuracy of grassland AGB during the peak growing season in arid areas. The specific objectives were as follows:

1) To classify vegetation and non-vegetation areas in UAV RGB images and extract vegetation location, geometric, and shape features.
2) To construct UAV-scale and satellite-scale AGB inversion models, respectively, and estimate and map grassland AGB during the peak growing season in the study area.
3) To compare the ground–satellite and ground–UAV–satellite modeling strategies and evaluate the contribution of intermediate-scale UAV observations to improving regional AGB inversion accuracy.

II. Study Area and Data

A. Study area

The study area is located in the northwestern part of the Xinjiang Uygur Autonomous Region, China, mainly including the Ili, Tacheng, and Altay regions. Geographically, it extends from 79°57′50″ to 87°57′46″ E and from 41°58′54″ to 47°58′51″ N (Fig. 2). The area is bounded by the Tianshan Mountains to the south and the Altay Mountains to the north and is located on the western side of the Junggar Basin. It is a key region for grassland animal husbandry research in Xinjiang. The elevation of the study area ranges from 172 to

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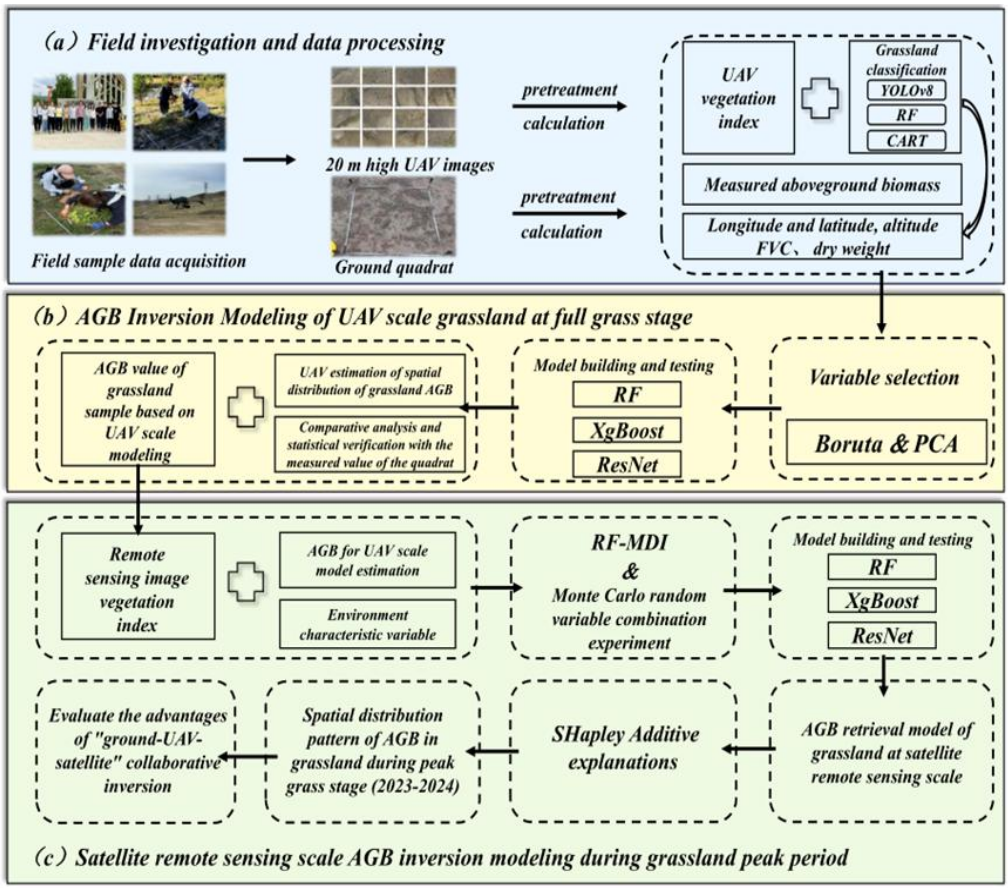


Fig. 1. Flow chart of this study.

6038 m, and the mean annual precipitation ranges from 131 to 1000 mm. The dominant soil types are chestnut soil and brown calcic soil [25]. The main grassland types in the study area include lowland meadow, mountain meadow, alpine meadow, temperate meadow steppe, temperate desert steppe, temperate steppe desert, and temperate desert [26]. Dominant plant species, such as Carex L., Festuca ovina, Kobresia bellardii, Alchemilla bungei, and Spiraea, are drought-tolerant, cold-resistant, and capable of strong soil stabilization, making them key species in the grassland ecosystems of the study area. This region is highly representative for studies on ecological patterns, responses to climate change, and grassland degradation.

B. Data Acquisition and Preprocessing

1) Field AGB Data Collection: Field surveys were conducted in July and August of 2023 and 2024, during which a total of 101 plots were established. Before plot establishment, stratified sampling was conducted by considering the topography, grassland distribution characteristics, and dominant vegetation types of the study area, so that the sampling points could cover different grassland types, elevation gradients, and vegetation growth conditions as much as possible. The dominant vegetation species and distribution of the sampling points in the study area are shown in Table A.I. Within each grassland type, relatively flat areas with relatively homogeneous vegetation distribution were preferentially selected for establishing 250 m × 250 m plots based on the topographic features and grassland distribution patterns. To reduce the influence of spatial autocorrelation on model training, the sampling points were spatially dispersed to avoid concentration in local areas. Meanwhile, under the premise of ensuring field survey safety, a certain distance was maintained between adjacent plots to reduce the influence of spatial clustering of sampling points on the model results. Five quadrats were established within each plot. The quadrat size was 1 m × 1 m for low herbaceous plots and 10 m × 10 m for herbaceous plots with tall shrubs. For each herbaceous plot containing tall shrubs, three additional 1 m × 1 m sub-quadrats were established. During sampling, herbaceous samples were collected by clipping at ground level, and their fresh weight was measured in the field. Meanwhile, information including plot longitude and latitude, elevation, grassland type, and vegetation coverage was recorded, and field photographs were taken. All samples were oven-dried at 60 °C for 48 h and then weighed to determine dry weight.

In addition, to further demonstrate the coverage of biomass variation across different grassland types, descriptive statistics of the measured AGB under different grassland types were calculated, including the mean, standard deviation, skewness, and distribution characteristics. The relevant results are shown in Table A.II.

2) UAV data: In this study, a DJI Mavic 3M Enterprise multispectral UAV equipped with multispectral cameras covering the green, red, red-edge, and near-infrared bands, a 20 MP visible-light camera, a three-axis gimbal, and an irradiance sensor was used to acquire images over 101 plots of 250 m × 250 m (Table A.III). During plot establishment,

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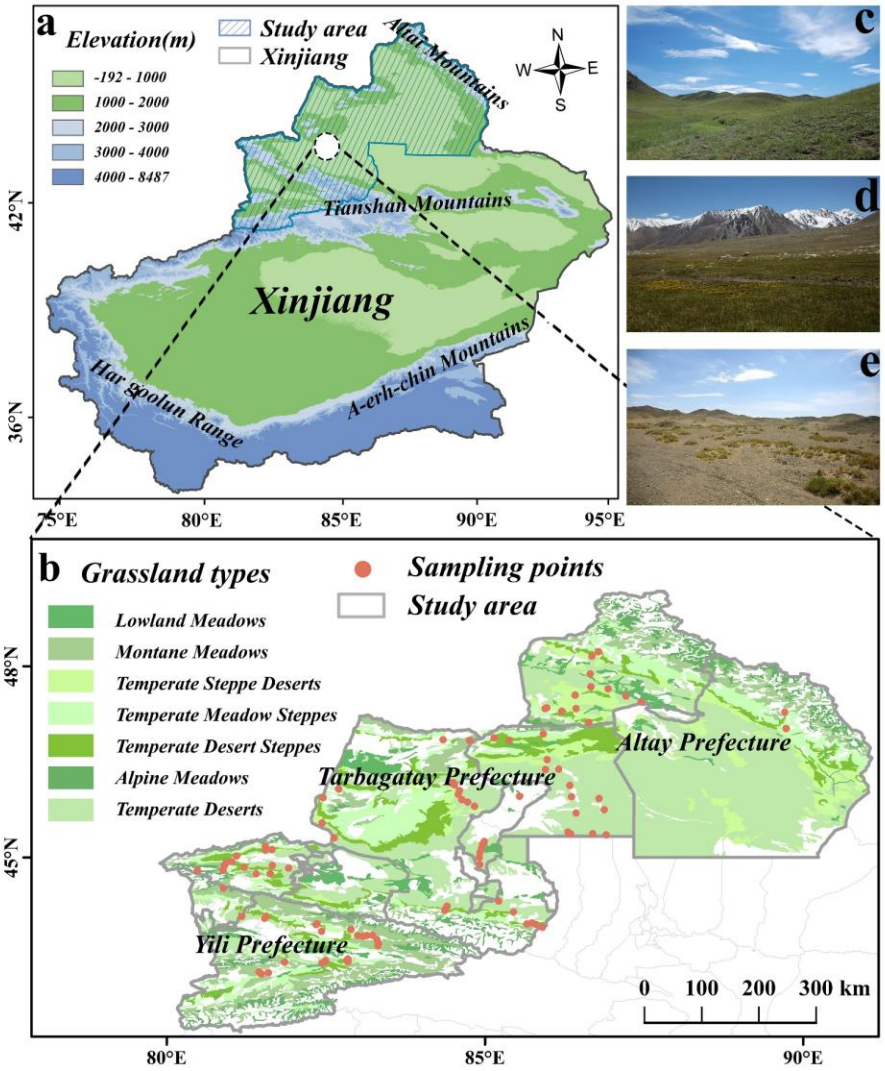


Fig. 2. Overview of the study area. (a) Location and elevation distribution of the study area. (b) Distribution of sampling points and grassland types. (c-e) Examples of grassland, meadow and desert landscapes.

relatively flat areas with relatively homogeneous vegetation distribution were preferentially selected as UAV sampling plots to reduce the influence of slope and terrain undulation on image acquisition and plot representativeness. Field UAV image acquisition was conducted as far as possible under clear and cloud-free conditions with sufficient illumination between 10:00 and 16:00.The flight routes were planned using the FragMAP aerial photography system [27]. According to the 250 m × 250 m plot scale, 16 UAV sampling positions were set within each plot. During image acquisition, the terrain-following flight mode was enabled throughout the flight. This mode dynamically adjusts the UAV flight altitude according to terrain undulation within the plot, allowing the UAV to maintain an approximately 20 m relative flight altitude above the ground at different terrain positions, thereby reducing the differences in relative flight altitude caused by slope [28]. The flight speed was 3 $m \cdot s^{-1}$, with a horizontal positioning accuracy of ±1.5 m and a vertical positioning accuracy of ±0.5 m. Multispectral images with a spatial resolution of approximately 1.0 cm were obtained. For each plot, 16 × 5 images were automatically acquired, including four multispectral images and one visible-light image at each sampling position. A single image covered an area of approximately 26 m × 35 m, and a total of 8080 images were obtained. The field data acquisition workflow is shown in Fig. 3.

The preprocessing of UAV multispectral images included radiometric calibration, irradiance correction, orthorectification, and quality control. First, the radiometric consistency calibration module of the UAV-mounted CMOS sensor was used to perform radiometric response consistency processing on images acquired within the same flight mission, in order to reduce the influence of sensor response differences on multispectral reflectance. Meanwhile, irradiance correction was performed based on the incident light information recorded by the UAV irradiance sensor to reduce the effects of changes in solar irradiance during flight on image brightness and reflectance. Subsequently, the original UAV images were orthorectified using DJI Terra. Specifically, the original images were imported into DJI Terra, and parameters such as

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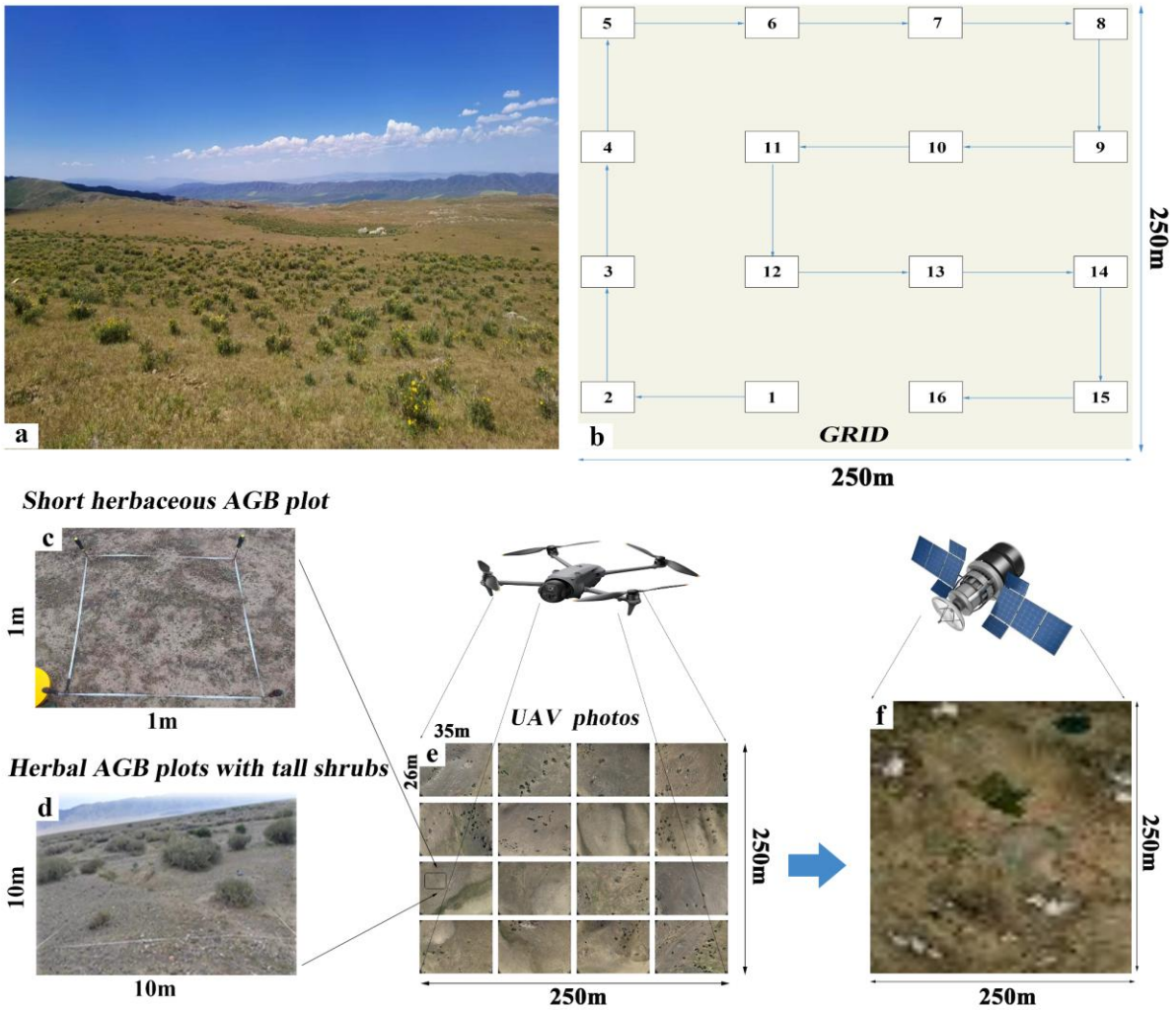


Fig. 3. Field data acquisition process. (a) Sample plot landscape. (b) Flight route of UAV in grid mode. (c, d) field quadrat collection. (e, f) Spatial scale comparison of UAV photos and satellite remote sensing data.

image POS data, output coordinate system, and spatial resolution were set. Two-dimensional reconstruction and orthorectification were then performed, and orthorectified images in GeoTIFF format were finally generated. Orthorectification is based on the transformation from central projection to orthographic projection and can reduce problems such as projection distortion and positional displacement caused by differences in shooting angle and terrain undulation under sloping terrain conditions [29]. Image quality control mainly included screening for exposure, blur, and shadows. Images with a saturated pixel proportion >5%, sharpness <0.6, or shadow coverage >10% were removed. After screening, a total of 7575 valid images were obtained.

The ground sampling distance (GSD) of the UAV multispectral images was calculated using the following equation:

$$\mathrm{GSD} = \frac{\mathrm{H}\times\mathrm{p}}{\mathrm{f}} \qquad (1)$$

where H is the relative flight altitude of the UAV, p is the sensor pixel size, and f is the actual focal length of the camera. In this study, the UAV flight altitude was 20 m, the actual focal length of the multispectral camera was 4 mm, and the pixel size was 2.0 μm; therefore, the GSD was 1.0 cm/pixel.

After preprocessing the UAV images, ArcGIS 10.8 was used to extract the mean reflectance of each band from the 16 imaging positions within each plot. The average value of these 16 positions was then calculated and used as the UAV-derived band reflectance for the 250 m × 250 m plot. Subsequently, based on these reflectance data, 14 vegetation indices were calculated using the Raster Calculator (Table A.IV) and used as spectral feature variables for the UAV-scale grassland AGB inversion model during the peak growing season.

3) Satellite data: The MODIS daily surface reflectance product MOD09GA (500 m) used in this study was obtained from NASA EOSDIS. MOD09GA is an atmospherically corrected surface reflectance product with a temporal resolution of 1 day. All data were downloaded from the Google Earth Engine (GEE) platform in TIFF format. According to the field survey and UAV sampling dates of each plot, the MOD09GA images acquired on the same day as the sampling date were selected to ensure temporal consistency among satellite remote sensing variables, ground measurements, and UAV observations. During preprocessing, scale factors were first applied to each reflectance band to convert the digital values into actual surface reflectance. Subsequently, the StateQA and QC_500m quality control

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bands were used to remove clouds, cloud shadows, and low-quality pixels, and abnormal reflectance values were also excluded to reduce the effects of cloud contamination and noisy pixels on subsequent vegetation index calculation. After quality control, the MOD09GA data were resampled to a spatial resolution of 250 m in GEE using bilinear interpolation to match the spatial resolution required for subsequent satellite-scale AGB modeling. Finally, 29 vegetation indices were calculated using the ArcGIS Raster Calculator (Table A.V) and used as spectral feature variables for satellite-scale grassland AGB inversion during the peak growing season.

4) Auxiliary data : The auxiliary data included meteorological, soil, and topographic factors. The meteorological data included monthly mean air temperature, monthly mean potential evapotranspiration, and monthly mean precipitation. These data were obtained from the MODIS land surface temperature product (MOD11A1), the MODIS evapotranspiration product (MOD16A2GF), and the 1901–2024 China 1 km monthly precipitation dataset provided by the National Earth System Science Data Center (NGSD), respectively. Based on the original monthly datasets, the multi-month mean air temperature and cumulative precipitation for the n months before sampling were calculated.

The soil data were obtained from the high-resolution soil property database released by the Land–Atmosphere Interaction Research Group of Sun Yat-sen University. These data included sand and clay contents at depths of 0–30 cm and 30–100 cm, as well as total phosphorus, total nitrogen, and organic matter contents at a depth of 23 cm. The topographic factors, including slope and aspect, were extracted from the SRTM DEM. To match the spatial resolution of the remote sensing data, all auxiliary data were resampled to 250 m using the nearest-neighbor method. Details are provided in Table A.VI.

III. Research Method

A. UAV-Based Grassland AGB Modeling

1) Classification Method: In addition to vegetation, bare soil and other background factors in the study area may also affect biomass inversion. To accurately extract vegetation regions and their morphological features, this study first classified vegetation and non-vegetation areas in the UAV images. For the three vegetation types, namely steppe, meadow, and desert, YOLOv8n-Seg, random forest (RF), and CART decision tree models were constructed separately, resulting in a total of nine models. The classification accuracy was evaluated using the F1-score and the area under the receiver operating characteristic curve (AUC).

YOLOv8n-Seg is an instance segmentation network in the YOLOv8 series. Its structure mainly consists of a feature extraction backbone, a feature fusion network or neck, and a segmentation head. The backbone is used to extract color, texture, boundary, and spatial structural features from UAV images. The neck integrates image features and improves the model's ability to identify vegetation patches of different sizes and irregular boundaries. The segmentation head outputs the class information, bounding boxes, and pixel-level segmentation results of vegetation targets. Before model training, vegetation classification labels were first generated based on the UAV images. Specifically, the images were converted to the HSV color space, and threshold ranges were set according to the green vegetation regions to preliminarily extract vegetation areas. Morphological processing was then applied to the preliminary extraction results to reduce scattered noise and enhance the continuity of vegetation regions. The contours of vegetation regions were subsequently extracted and converted into the YOLO segmentation label format. After model training, the trained YOLOv8n-Seg model was applied to segment vegetation in the UAV images, thereby obtaining pixel-level vegetation segmentation masks corresponding to the original images [30]. RF and CART were used as traditional classification methods for comparative analysis. Both methods used manually annotated binary vegetation masks as labels and UAV image pixel features as input variables to construct binary classification models for vegetation and non-vegetation areas. RF performs classification by constructing multiple decision trees and integrating their classification results, which can reduce the sensitivity of a single decision tree to sample perturbations. CART constructs a binary decision tree by recursively partitioning the feature space to classify vegetation and non-vegetation categories. The sample data were divided into training and testing sets at a ratio of 7:3.

Based on the F1-score and AUC evaluation results, the model with the highest accuracy among the three classification methods was selected as the final vegetation extraction model. Subsequently, the pixel-level vegetation segmentation masks or vegetation/non-vegetation classification results output by this model were converted into binary vegetation masks, in which vegetation areas were assigned a value of 1 and non-vegetation areas were assigned a value of 0. Based on the binary masks, the location, shape, and geometric features of vegetation patches were further extracted for subsequent biomass inversion. The key parameter settings of the three classification models are presented in Table A.VII, and detailed information on the extracted features is provided in Table A.VIII. The principles and characteristics of the three classifiers are described in detail in the Supplementary Materials.

2) Feature Selection Based on Boruta and PCA: Before grassland AGB modeling, a feature set was constructed using multispectral UAV data, and feature values were extracted within the herbaceous quadrats. A total of 25 features were extracted from the UAV images, including 14 spectral features, two vegetation features, three location features, three shape features, and three geometric features (Table A.IX). To alleviate multicollinearity caused by high-dimensional features and the limited number of ground herbaceous quadrats, the Boruta algorithm and principal component analysis (PCA) were used for variable selection and dimensionality reduction, respectively. Before modeling, all variables were standardized using Z-score normalization, and highly redundant and collinear variables were excluded based on Pearson/Spearman correlation tests ($|r| > 0.8$) and the variance inflation factor ($VIF > 5$). The Boruta algorithm identifies important variables that make significant contributions to the model by comparing the Z-scores of the original features with those of randomly

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generated shadow features [31]. In this study, the Boruta algorithm was implemented in R software (version 4.3.2) using the "Boruta" package [32]. PCA analyzes the structure of multidimensional features by examining correlations among variables and transforms multiple original variables into a smaller number of comprehensive variables, thereby achieving dimensionality reduction [33]. PCA was performed using SPSS 25 software [34]. Subsequently, RF, XGBoost, and ResNet models were constructed based on the variable selection results of Boruta and PCA, respectively, resulting in six model combinations. The predictive performance of each model under different variable selection methods was then compared. The detailed principles of the variable selection methods are provided in the Supplementary Materials.

3) UAV-Scale AGB Modeling: A ground–UAV-scale grassland AGB inversion model was constructed based on the ground-measured AGB data and UAV image features. To match the resampled MOD09GA pixel scale (250 m × 250 m), candidate resampled MOD09GA pixels suitable for ground sampling were selected before field investigation, considering grassland type distribution, topography, and road accessibility. The boundaries, four corner coordinates, and centroid locations of these resampled pixels were then extracted, and 250 m × 250 m plots were established based on the centroid and boundary of each selected pixel [35]. The plot boundary was also used as the 250 m × 250 m UAV observation extent for flight-route planning and image acquisition, ensuring spatial consistency among the plot extent, UAV observation extent, and resampled MODIS pixels.

Within each 250 m × 250 m plot, five ground quadrats were established using a five-point sampling method, and the mean AGB of the five quadrats was calculated as the measured AGB value of the plot [36], [37]. Based on the UAV multispectral images acquired within each plot, the mean reflectance of image pixels in each band was extracted, and vegetation index features were calculated. Meanwhile, vegetation geometric and shape features were extracted based on the vegetation classification results derived from UAV visible-light images. Finally, the UAV-scale extracted features were used as independent variables, and the plot-level measured AGB was used as the dependent variable to construct the UAV-scale AGB inversion model. The detailed principles of the model algorithms are provided in the Supplementary Materials. All models were implemented in a Python environment, and five-fold cross-validation was used to evaluate model stability and generalization ability [38]. Model accuracy was assessed using two statistical indicators: the coefficient of determination (R^2) and the root mean square error (RMSE). A higher R^2 value and a lower RMSE value indicate higher model accuracy and stability [39].

The formulas for R^2 and RMSE are as follows:

$$R^2=1-\frac{\sum_{i=1}^{n}(Z_i-Z_y)^2}{\sum_{i=1}^{n}(Z_i-Z_x)^2} \quad (2)$$

$$RMSE=\sqrt{\frac{\sum_{i=1}^{n}(Z_x-Z_y)^2}{n}} \quad (3)$$

where Z_x is the observed value, Z_y is the predicted value, Z_i represents the i-th estimated value, and n is the number of validation samples.

B. Satellite-Based AGB Modeling

1) Variable Combination Selection Based on the Monte Carlo Method: In this study, the Random Forest Mean Decrease Impurity (RF-MDI) method was used to evaluate the importance of 79 variables (Table A.X) and the relative importance of each variable within each category [40]. Considering the interactions caused by collinearity among variables in AGB modeling, Monte Carlo simulations driven by different combinations of the 79 variables were used to evaluate the performance of different algorithms.Variable combination experiments were first conducted within each category, yielding a total of 10,058 combinations, including 30 combinations of climatic factors, 14 combinations of topographic factors and soil properties, and 10,000 combinations randomly selected from 29 satellite optical indicators based on random approximation theory [41]. Based on the minimum mean absolute error (MAE) loss, the optimal variable combination within each of the four categories was selected, and 3,600 final inter-category combinations were generated. For each combination, RF, XGBoost, and ResNet algorithms were run to calculate the MAE loss for model evaluation [42].

The MAE was calculated as follows:

$$MAE=\frac{1}{n}\sum_{i=1}^{n}\left|Z_x-Z_y\right| \quad (4)$$

where Z_x and Z_y are the observed and predicted AGB values, respectively; Z_i is the model estimate for the i-th sample; and n is the number of validation samples.

2) Satellite-Based AGB Modeling: Based on the grassland AGB inversion results obtained at the UAV scale, a satellite-scale grassland AGB inversion model was further constructed. The original spatial resolution of MOD09GA is 500 m, and it was resampled to 250 m in GEE. Accordingly, the UAV sampling plots were set to 250 m × 250 m. The centroids of the previously selected 250 m × 250 m MODIS pixels were used as the plot centers, and the pixel boundaries were used to define the plot extent. UAV flight routes and observation extents were then planned accordingly, so that the UAV plots were consistent with the corresponding MODIS pixels in both spatial location and boundary. Subsequently, remote sensing variables were extracted from the 250 m MODIS pixels whose spatial extents were consistent with those of the UAV plots, and the UAV-scale AGB inversion result of each plot was matched with the remote sensing variables of the corresponding pixel. Meanwhile, auxiliary variables, including meteorological, topographic, and soil variables, were unified to a spatial resolution of 250 m. Finally, the UAV-scale AGB inversion results were used as the dependent variable, while the remote sensing variables and auxiliary variables of the corresponding MODIS pixels were used as independent variables. The optimal factor set was determined through Monte Carlo variable-combination selection, and ResNet, RF, and XGBoost models were then used to construct satellite-scale grassland AGB inversion models. Model accuracy was evaluated using R^2 and RMSE. Finally, the best-performing model was used to

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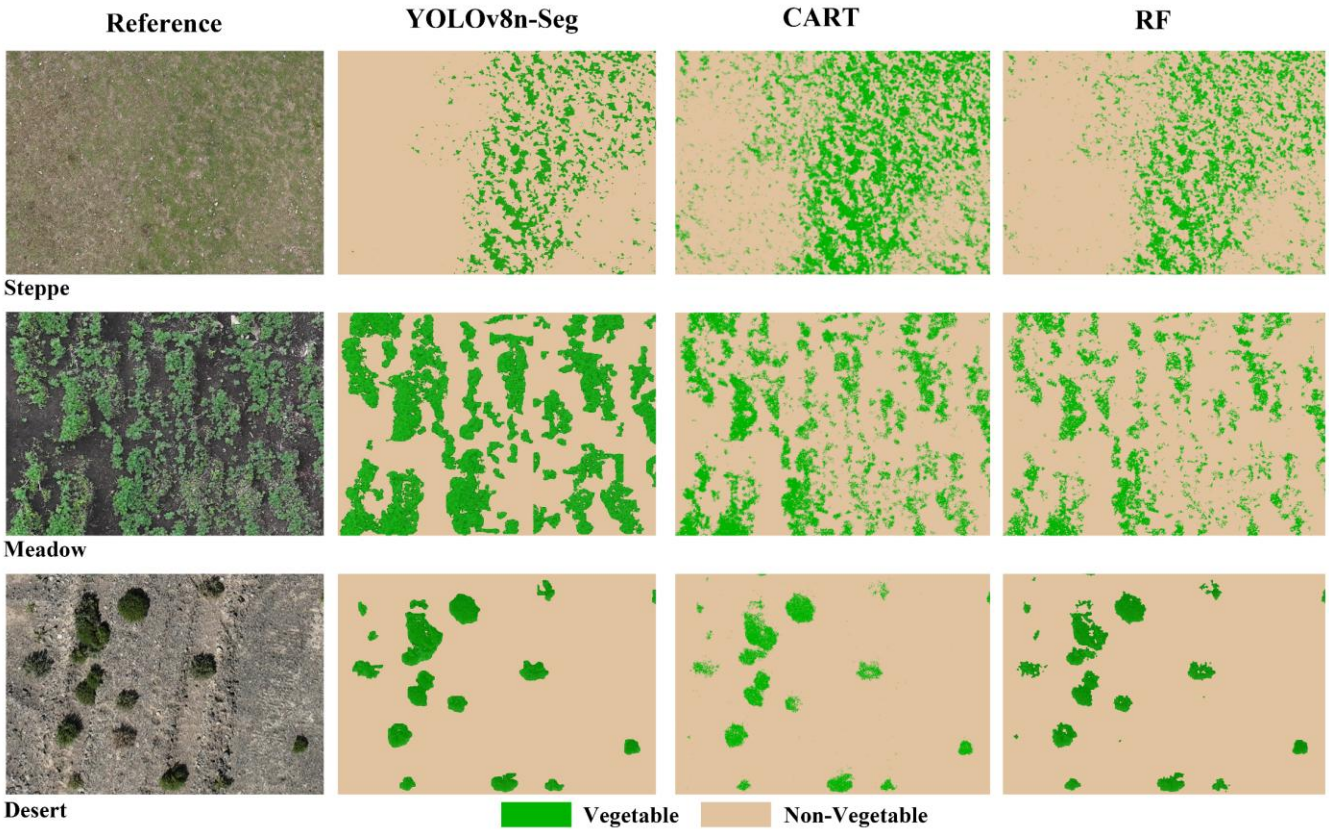


Fig. 4. UAV-scale classification results for the three vegetation types.

generate the spatial distribution map of grassland AGB at the satellite scale, thereby achieving the spatial upscaling of grassland AGB from the UAV scale to the satellite scale.

Through this approach, UAV data served as a scale bridge between ground-plot observations and MODIS satellite data, avoiding the scale inconsistency caused by directly matching small-scale ground-plot AGB with relatively coarse-resolution satellite variables [43]. To analyze the matching errors between UAV-scale and satellite-scale AGB results, this study calculated the differences among AGB inversion results at different scales and used boxplots to show the error distribution characteristics (Fig. A4).

3) Shapley Additive Explanation for the AGB Model: To accurately interpret the best-performing grassland AGB inversion model during the peak growing season in the study area, SHapley Additive exPlanations (SHAP) values were used for model interpretation. SHAP is one of the most advanced methods for interpreting the results of learning models. Based on game theory, SHAP calculates feature importance by estimating the average marginal contribution of each feature to the model output [44]. In this study, SHAP values were used to generate summary plots and dependence plots. These results helped identify the influence of each feature value on the model output, clarify how individual features contributed to driving model predictions, and evaluate the effect of each feature on model prediction results. SHAP values were calculated in a Python environment using the "shap" package [45].

C. Evaluating the Advantages of Ground–UAV–Satellite Upscaling

In traditional grassland AGB modeling, ground-quadrat AGB data are usually used directly as modeling samples. To explore the advantages of the ground–UAV–satellite upscaling approach, this study combined the AGB data from 101 ground herbaceous quadrats with satellite images to establish a ground–satellite-scale AGB estimation model. In addition, by integrating UAV images with satellite images, the AGB data derived from the ground–UAV-scale inversion were used as the dependent variable to construct a ground–UAV–satellite-scale AGB estimation model [46]. Five-fold cross-validation was used for accuracy validation in both modeling pathways. To further evaluate the spatial inversion errors of the two pathways, the trained ResNet model was used for spatial inversion to generate spatial error maps of AGB, thereby comparing the biomass estimation errors of the two pathways at the spatial scale.

IV. Results

A. UAV-Scale Grassland AGB Inversion Model

1) Classification Results and Accuracy Assessment: Table I shows the classification results of different classifiers, including YOLOv8n-Seg, RF, and CART, for distinguishing vegetation from bare soil and other non-vegetation factors across three grassland types: steppe, meadow, and desert. The results showed that YOLOv8n-Seg outperformed RF and CART across all three grassland types, with the highest F1-score reaching 0.800. This indicates that YOLOv8n-Seg can

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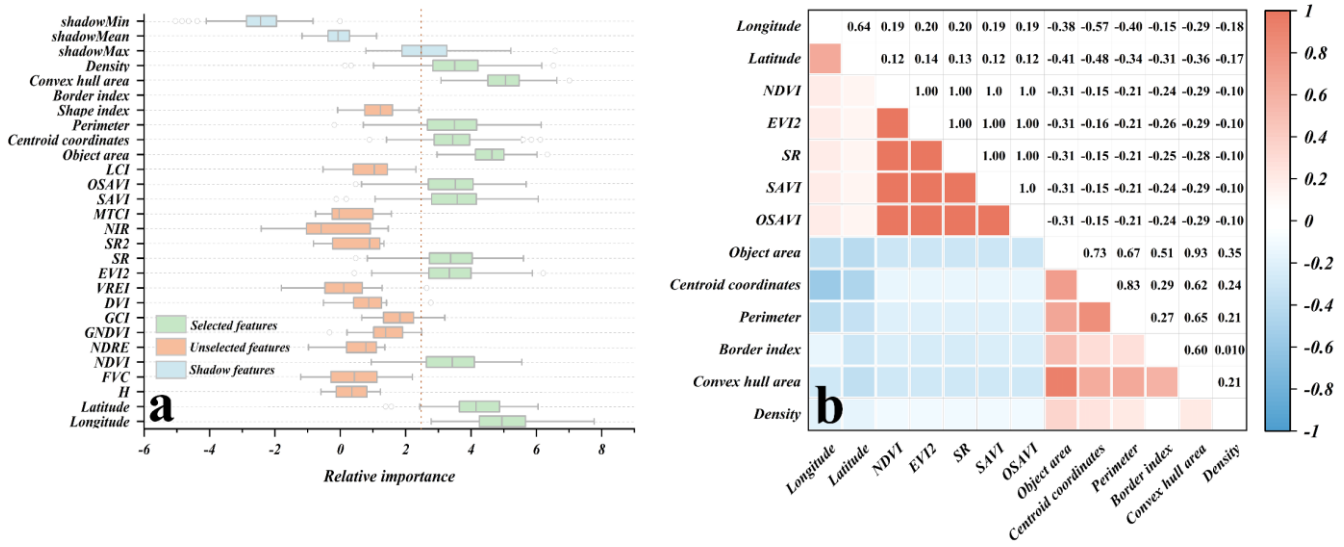


Fig. 5. Boruta algorithm-based variable selection process for the UAV-scale model. (a) Variable importance results. (b) Collinearity assessment among high-importance variables (variables with VIF > 5 were excluded; $p < 0.001$).

accurately classify and identify vegetation, bare soil, and other non-vegetation factors in the study area. Therefore, YOLOv8n-Seg was ultimately selected to classify vegetation in all UAV images of the study area and to extract the corresponding features. As shown in the classification results in Fig. 4, the classification results of YOLOv8n-Seg showed high spatial consistency with the reference images across the steppe, meadow, and desert grassland types, and effectively reflected the spatial distribution characteristics of vegetation patches. By contrast, RF and CART showed poorer vegetation patch continuity or less distinct boundaries in some areas. Overall, YOLOv8n-Seg demonstrated good classification stability across different grassland types.

TABLE I

TITLE STATISTICS OF CLASSIFICATION ACCURACY IN THE STUDY AREA

Ecosystem Type	Method	F1-Score	AUC
Steppe	YOLO v8n-Seg	0.800	0.900
	RF	0.418	0.752
	CART	0.520	0.731
Meadow	YOLO v8n-Seg	0.787	0.999
	RF	0.752	0.816
	CART	0.765	0.817
Desert	YOLO v8n-Seg	0.761	0.896
	RF	0.546	0.817
	CART	0.543	0.777

2) UAV-based variable selection results: At the UAV scale, PCA and the Boruta algorithm were used to select the variables after vegetation classification. A total of 25 candidate variables were initially included, covering five categories: spectral features, vegetation features, shape features, geometric features, and location features. Based on the results of the Boruta algorithm, Longitude, Latitude, SAVI, Perimeter, Border index, Convex hull area, and Density were identified as model independent variables selected by Boruta (Fig. 5). According to the PCA variable selection results, the variables with a cumulative contribution rate exceeding 85%, including DVI, MTCI, Centroid coordinates, GCI, Border index, Latitude, and FVC, were selected as model independent variables based on PCA (Table II).

TABLE II

PCA PRINCIPAL COMPONENT ANALYSIS VARIABLE SCREENING RESULTS

Model type	Main components	Maximum importance variable of each component	Accumulated contribution%
UAV Scale	PCA1	DVI	87.58
	PCA2	MTCI	
	PCA3	Centroid coordinates	
	PCA4	GCI	
	PCA5	Border index	
	PCA6	Latitude	
	PCA7	FVC	

3) Comparison of Model Performance and Accuracy Before and After Classification: Based on the two variable selection strategies, a total of 12 UAV-scale AGB inversion models were constructed before and after vegetation classification using ResNet, RF, and XGBoost, respectively (Fig. 6). In terms of model performance before and after classification, the test-set R^2 of the AGB inversion models before classification ranged from 0.56 to 0.78, and the RMSE ranged from 179.62 to 262.22 kg/ha. After classification, the test-set R^2 increased significantly to 0.62–0.85, while the RMSE decreased to 149.24–246.04 kg/ha. These results indicate that vegetation classification significantly improved the prediction accuracy of the AGB inversion models. This suggests that, at the UAV scale, when geometric and shape features were extracted only from vegetation areas, these features could more accurately reflect vegetation structural characteristics and growth

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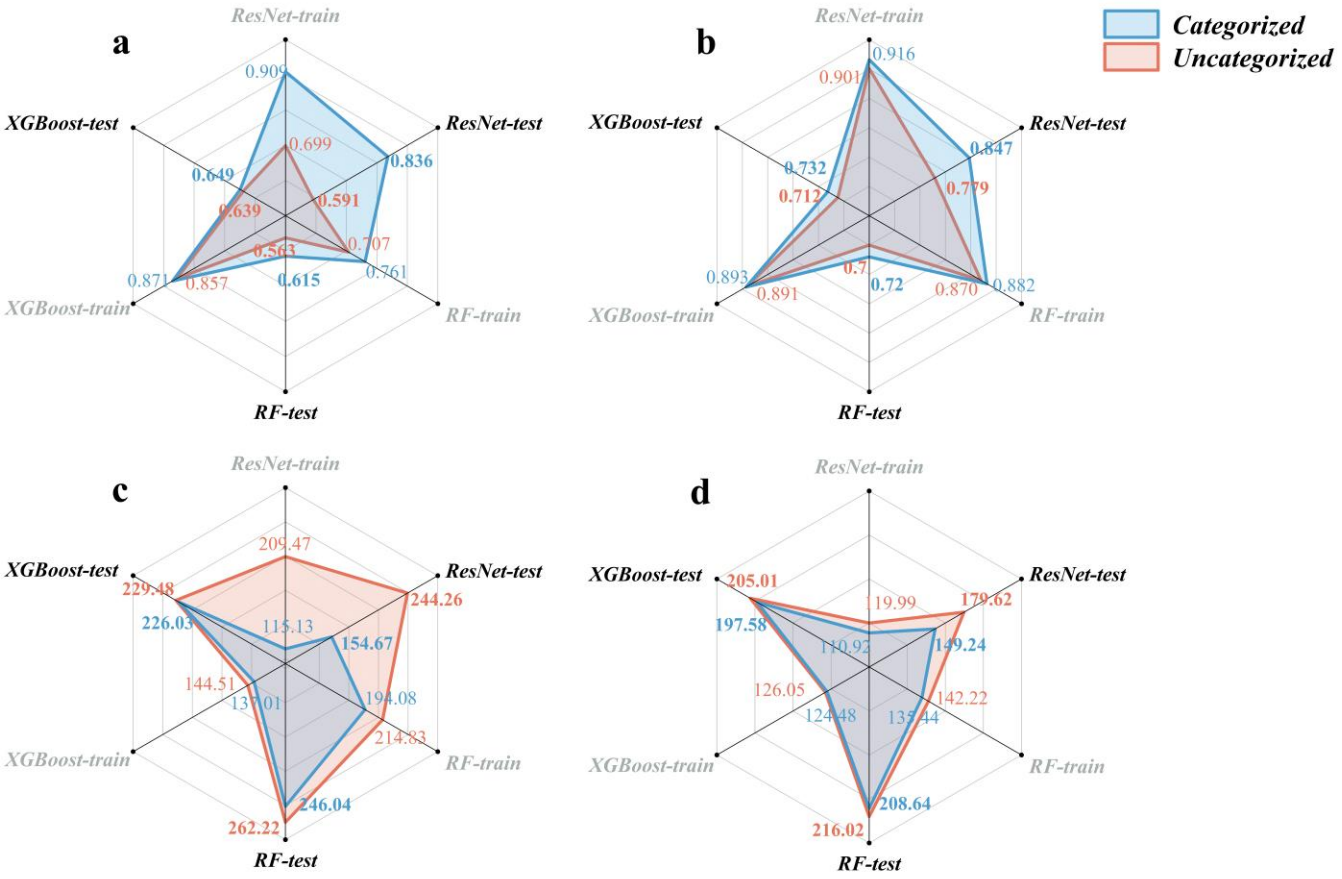


Fig. 6. UAV-scale model training accuracy before and after classification. (a, c) Models constructed using variables selected by the Boruta algorithm. (b, d) Models constructed using variables selected by PCA. (a, b) Use the coefficient of determination (R^2). (c, d) Use the root mean square error (RMSE, kg/ha).

conditions, thereby enhancing the ability of the models to predict grassland AGB.

In terms of variable selection, among the models constructed using variables selected by the Boruta algorithm, ResNet achieved the highest overall inversion accuracy, followed by XGBoost and RF. For the models constructed using variables selected by PCA, ResNet also showed the best performance. In addition, the biomass inversion models constructed using PCA-selected variables consistently outperformed those based on the Boruta algorithm. Considering the performance of different variable selection methods and model combinations, the post-classification ResNet biomass inversion model constructed using PCA-selected variables performed best, with a test-set R^2 of 0.85 and an RMSE of 149.24 kg/ha. Therefore, this model was regarded as the optimal model for UAV-scale grassland AGB inversion in this study. A comparison between the inversion results and measured data showed that the grassland AGB estimated by the optimal model ranged from 80.53 to 2539.25 kg/ha, with a mean and median of 360.57 kg/ha and 245.1 kg/ha, respectively. The field-measured AGB values ranged from 90 to 2400 kg/ha, with a mean and median of 376.53 kg/ha and 250 kg/ha, respectively (Fig. A1).

B. Grassland AGB Inversion Model at the Satellite Scale

1) Importance of Selected Variables: According to the RF-MDI importance ranking of the 79 variables (Fig. 7b), geographic factors showed the greatest importance in AGB estimation, contributing 68%, whereas the contributions of soil properties, spectral indicators, and climatic factors were all ≤16%. Notably, among the 79 variables, Longitude (25%), DEM (20%), Slope (11%), and Latitude (5%) jointly contributed 61% to the spatial variation in AGB. Among the remaining variables, RTP (3%) and NLI (3%) showed relatively high importance rankings (Fig. 7a).

Based on the importance rankings of predictors within each specific category (Fig. 7c), Longitude, NLI, RTP, and tmp8-5 maintained robust contributions, with importance values of 37%, 24%, 19%, and 7%, respectively. Consistently, the geographic factor Slope contributed more than 30% to AGB estimation in the XGBoost, RF, and ResNet models selected using the Monte Carlo approach (Fig. 7d). Soil properties and spectral indicators contributed more than 18% and 15%, respectively. By contrast, climatic factors showed relatively small contributions (<10%), except that the ResNet model assigned higher importance scores to the climatic factors pre8-1 and tmp8-5.

2) Comparison of Model Results and Accuracy: Based on the 79 variables and the predictors selected by Monte Carlo

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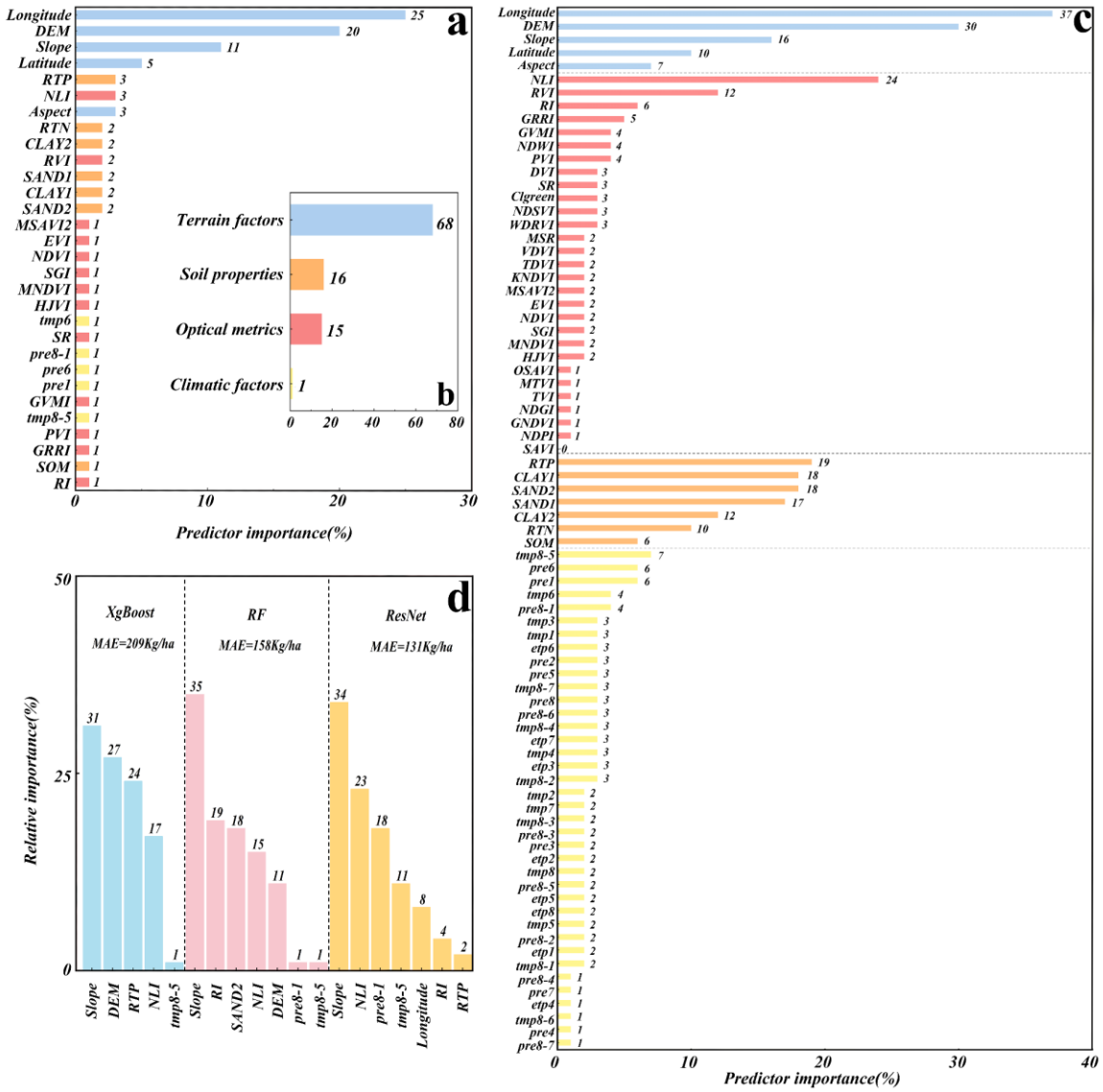


Fig. 7. Predictor importance for the satellite-scale grassland AGB model. (a) Importance of the top 30 driving variables across four categories. (b) Overall importance of all 79 driving variables summarized by the four major categories. (c) Distribution of predictor importance within each category. (d) Minimum MAE loss and predictor importance for the three algorithms obtained through Monte Carlo simulation.

screening, six learning models were constructed using ResNet, RF, and XGBoost, respectively (Fig. 8). Overall, the test-set R^2 values of the six models ranged from 0.58 to 0.72, and the RMSE values ranged from 202.15 to 260.62 kg/ha. Compared with the models constructed using all 79 variables (R^2 = 0.58–0.68), the models based on Monte Carlo-selected predictors showed slightly improved performance, with R^2 values ranging from 0.60 to 0.72.

Specifically, the AGB values inverted by the ResNet, RF, and XGBoost models were consistently underestimated, whereas Monte Carlo screening reduced part of the systematic bias and improved prediction accuracy. The Monte Carlo screening results indicated that using all 79 variables did not improve the AGB modeling performance of the algorithms.

3) Interpretation of the Grassland AGB Model: In this study, SHAP values were used to interpret the AGB model with the highest R^2. The summary plots (Fig. 9a and Fig. 9b) show the SHAP values for each sample, and the feature importance was ranked according to the mean absolute SHAP value. For the optimal AGB model, NLI had the greatest influence on the model output, with higher NLI values corresponding to higher SHAP values. Geographic factors also had significant effects on the AGB model, with Longitude and Slope ranking second and third in terms of model importance, respectively. The dependence plots in Fig. 9c, Fig. 9d, Fig. 9e, and Fig. 9f further illustrate the effects of important features in the model. The relationship between NLI and AGB showed a certain nonlinear pattern. When Longitude in the AGB model ranged from 83° to 85°, the SHAP values were mostly negative, indicating lower estimated AGB values. Slope showed a clear negative effect; when the slope was small, the SHAP values were close to zero or slightly positive. The SHAP variation of RTP was relatively gentle, showing only a slight increasing trend.

4) Spatial Distribution of Grassland AGB During the Peak Growing Season of 2023–2024 : From 2023 to 2024, grassland AGB in the study area ranged from 70.46 to 5000

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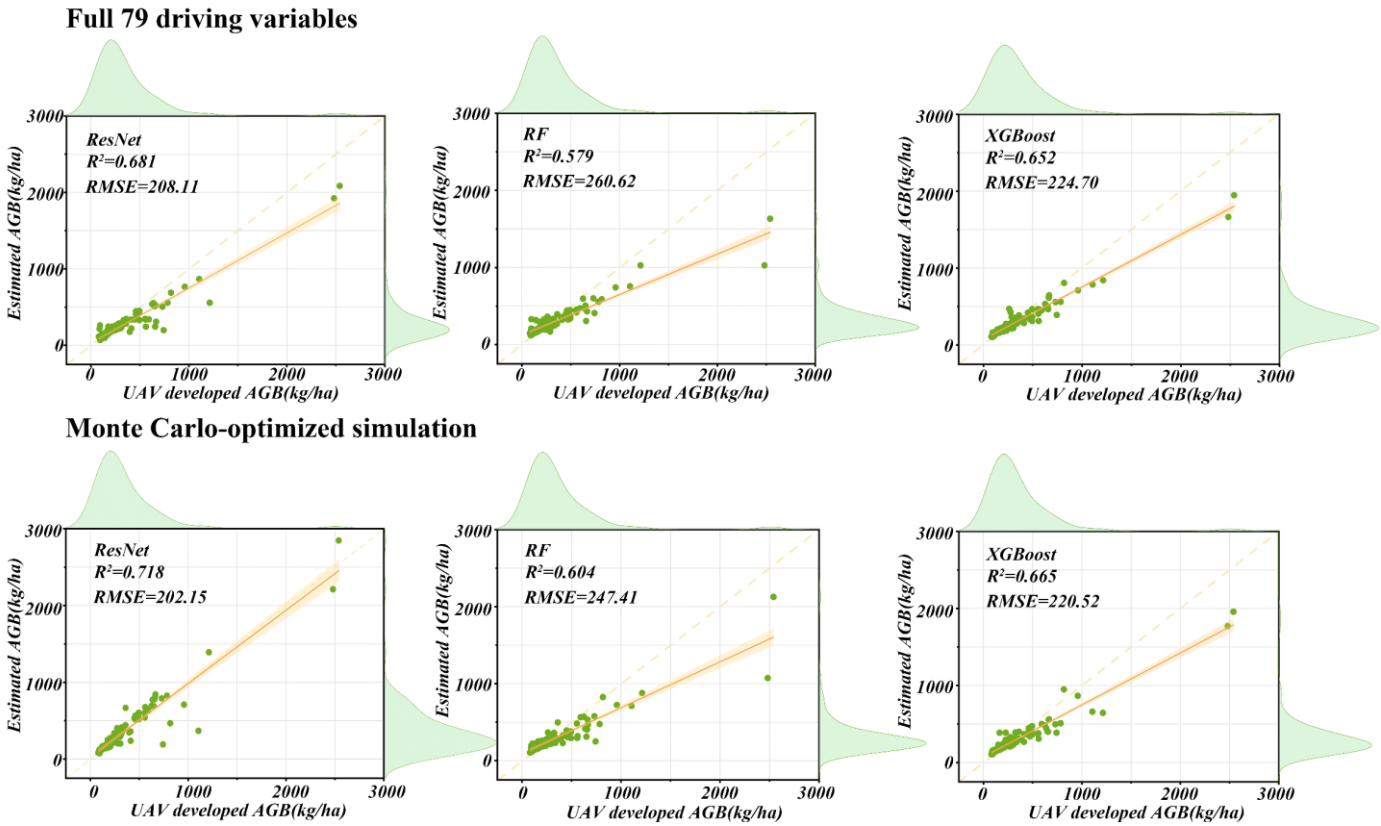


Fig. 8. Comparison of model performance at the satellite-scale for grassland AGB estimation using all 79 driving variables versus Monte Carlo–optimized predictors.

kg/ha during the peak growing season, with areas of higher grassland biomass mainly concentrated in the northeastern and southeastern parts of the study area (Fig. 10a). Specifically, areas with grassland AGB values below 700 kg/ha accounted for 43.2% of the total area, those with AGB values ranging from 700 to 2525 kg/ha accounted for 45.6%, and areas with relatively high grassland AGB values (>2525 kg/ha) accounted for approximately 11.2%. Overall, grassland AGB showed an increasing trend with elevation. The mean grassland AGB reached its lowest value of 1023 kg/ha at elevations of 207–731 m and its highest value of 1712 kg/ha at elevations of 2360–3366 m (Fig. 10b). In terms of AGB distribution among different grassland types, meadow grasslands had markedly higher mean AGB values, whereas desert grasslands had relatively lower mean AGB values (Fig. 10c).

C. Advantages of Ground-UAV-Satellite Upscaling in AGB Estimation

Based on the five-fold cross-validation results of the ground–satellite method and the ground–UAV–satellite upscaling method (Fig. 11c and Fig. 11d), the upscaling method developed in this study showed better performance. By introducing UAV samples to effectively bridge the scale gap, this method improved the R^2 to 0.718 and reduced the RMSE to 202.15 kg/ha compared with the ground–satellite method.

In terms of the spatial distribution of grassland AGB inversion errors obtained using the two methods (Fig. 11a and Fig. 11b), the biomass errors of the ground–satellite method were <600 kg/ha in 38.5% of the area and ranged from 600 to 1200 kg/ha in 34.3% of the area, while relatively large errors (>2000 kg/ha) accounted for approximately 8.4%. By contrast, the biomass errors of the ground–UAV–satellite upscaling method were more concentrated, with 84% of the area showing errors below 600 kg/ha. Among them, 45.1% of the area had errors below 200 kg/ha, and 36.6% of the area had errors between 200 and 400 kg/ha.

Overall, the ground–UAV–satellite upscaling method outperformed the ground–satellite method in terms of both grassland biomass inversion accuracy and error distribution, indicating that the introduction of UAV data can effectively bridge the scale gap in traditional ground–satellite-based biomass estimation.

V. Discussion

A. Mechanisms Underlying Improved Grassland AGB Inversion Accuracy Using YOLOv8n-Seg Based Vegetation Classification

In this study, vegetation and non-vegetation objects such as bare soil in UAV images were classified, and the accuracy of AGB inversion models before and after classification was compared. The results showed that the vegetation-classification-based model effectively improved the accuracy of grassland AGB estimation in northwestern Xinjiang. Specifically, grasslands contain vegetation, bare soil, and other non-vegetation objects, which are often highly mixed at the UAV scale [47]. If the original images are directly used to

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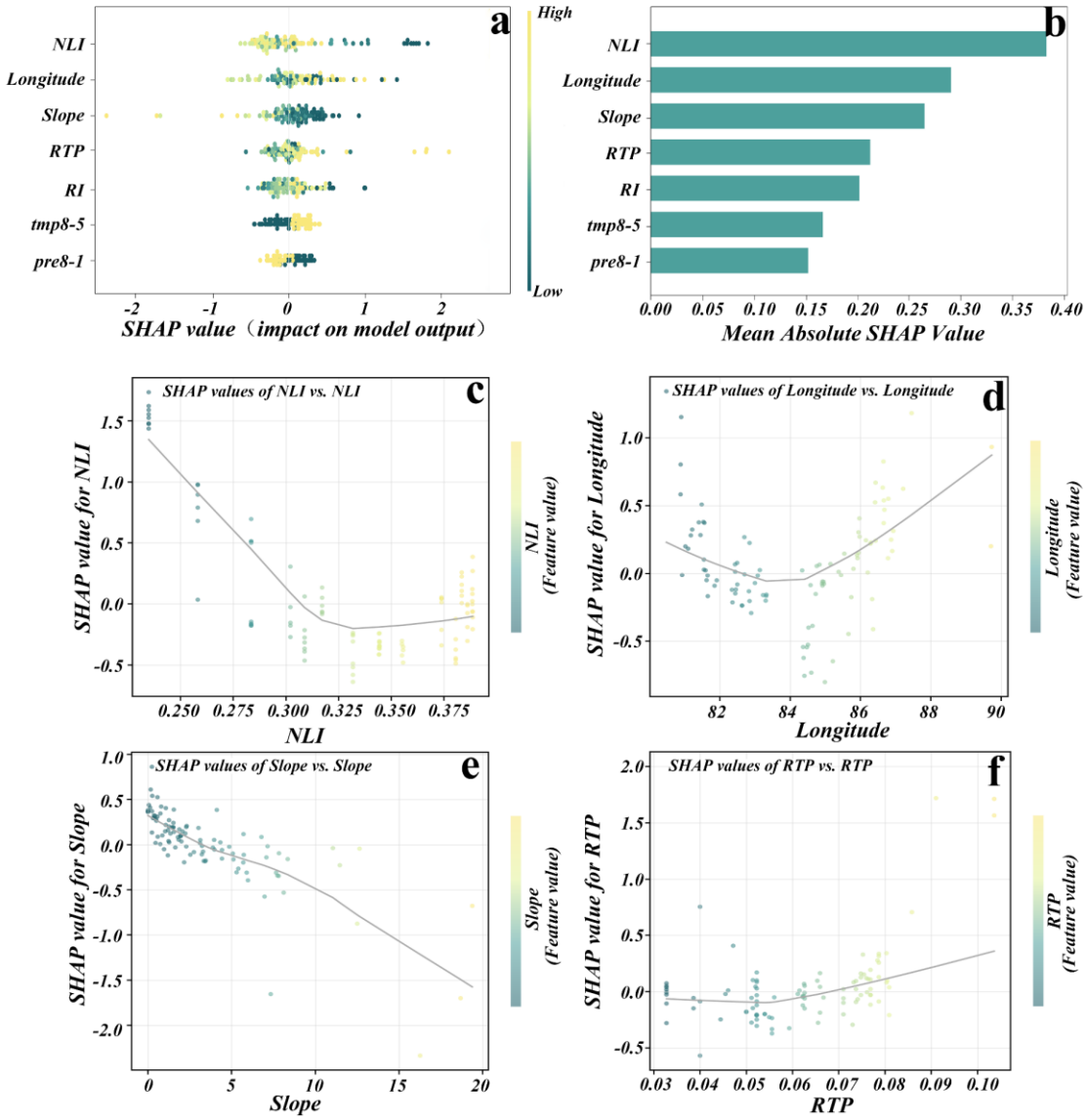


Fig. 9. SHAP analysis of the optimal grassland AGB model. (a) Summary plot of the optimal AGB model. (b) Mean absolute SHAP importance of all variables. (c-f) Dependence plots illustrating the effects of key predictors on model output.

construct the AGB inversion model, non-vegetation information such as bare soil inevitably participates in the modeling process, thereby interfering with the relationship between vegetation spectral features and biomass.

For the fine classification of vegetation and non-vegetation areas, the choice of classification method has a critical influence on classification accuracy and subsequent AGB inversion performance. In recent years, deep learning-based classification methods have shown outstanding performance in high-resolution UAV image processing. As an advanced deep learning model, YOLOv8n-Seg generally outperforms traditional machine learning and earlier deep learning methods in terms of feature extraction capability, classification accuracy, and adaptability to complex backgrounds [48]. In this study, the F1-score of YOLOv8n-Seg was significantly higher than those of RF and CART in some grassland types. This difference was mainly attributed to the different capabilities of the models in utilizing spectral, spatial, and texture information. RF and CART are traditional pixel-level classification methods that mainly distinguish classes based on spectral differences among individual pixels or local pixels, and their ability to use information on the spatial structure, texture variation, and boundary morphology of vegetation patches is limited. In UAV images of arid grasslands, vegetation is usually sparsely and patchily distributed, and non-vegetation backgrounds such as bare soil, litter, and shadows may show spectral similarity to vegetation. Therefore, relying only on spectral information can easily lead to confusion between vegetation and non-vegetation areas. In particular, in vegetation edges, small-scale patches, and sparsely vegetated areas, RF and CART are prone to problems such as unclear boundary delineation, omission of scattered vegetation, or fragmentation of continuous vegetation into broken pixels. By contrast, YOLOv8n-Seg can comprehensively learn spectral, texture, and morphological features from images through deep convolutional networks,

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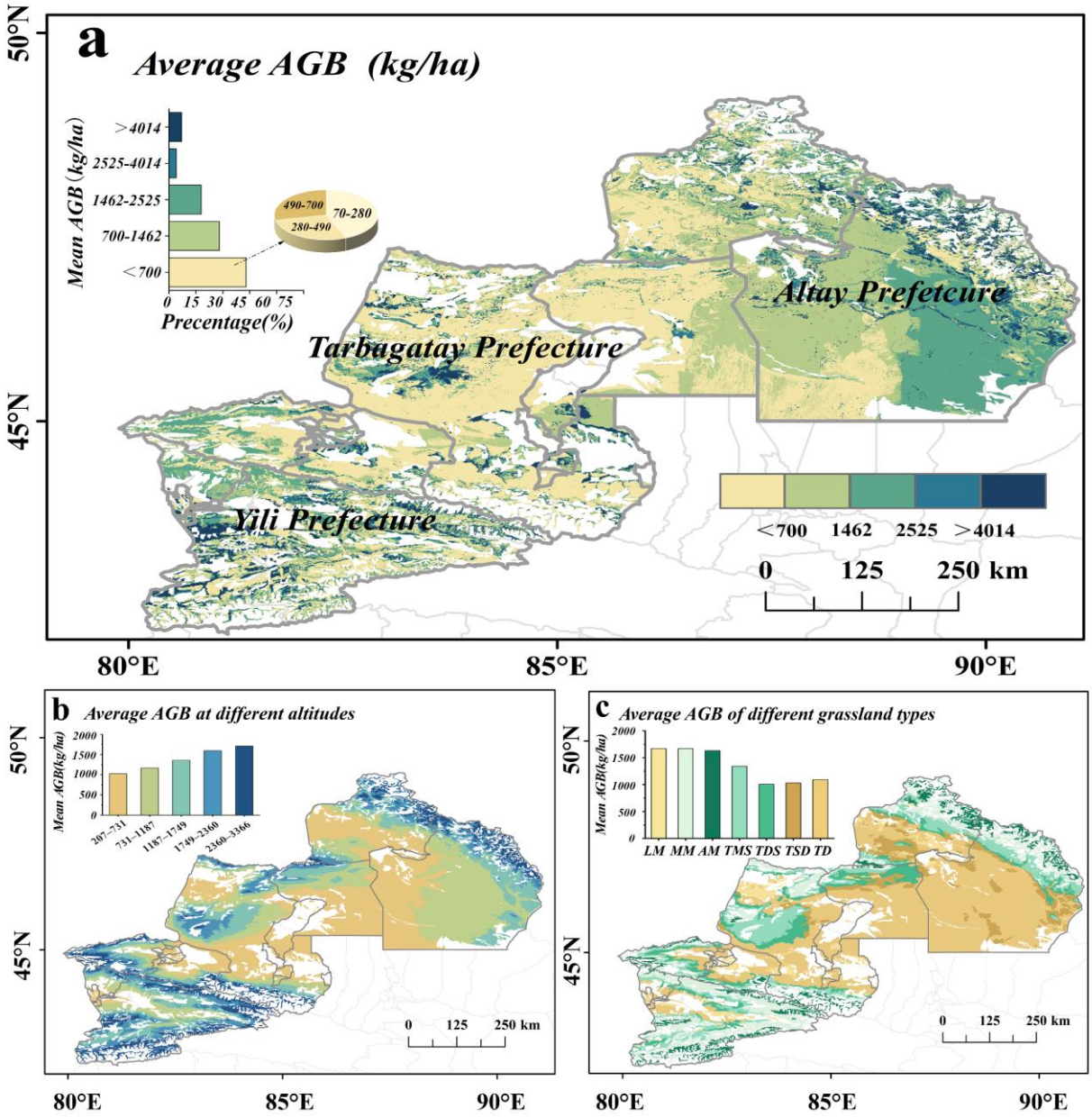


Fig. 10. (a) Spatial distribution of annual mean grassland AGB during the peak growing season in the study area. (b) and (c) spatial distributions of annual mean grassland AGB under different elevation ranges and different grassland types, respectively.

thereby identifying vegetation regions more accurately under complex background conditions [49]. Across different grassland types, this model effectively maintained the spatial continuity and integrity of vegetation patches. In particular, in desert grasslands with sparse vegetation distribution and small patch sizes, it could still effectively identify scattered vegetation patches and reduce omission errors. In addition, in transition zones between vegetation and bare soil, YOLOv8n-Seg showed stronger capability in characterizing the spatial and texture features of vegetation patches, making vegetation extraction results more accurate and stable. These advantages not only improved the classification accuracy of vegetation and non-vegetation areas, but also provided a more reliable basis for removing background information, extracting vegetation geometric and shape features, and constructing post-classification AGB inversion models.

Based on the above classification results, background information such as bare soil was removed before modeling, and a model focusing only on vegetation-covered areas was constructed, thereby providing a necessary prerequisite for extracting vegetation geometric and shape features. Subsequently, variable selection was performed on the extracted features, and two types of features, Border index and Centroid coordinates, were finally retained. This indicates that the boundary complexity and spatial distribution characteristics of vegetation patches contain important structural information related to grassland AGB [50]. Among them, Border index characterizes the regularity and

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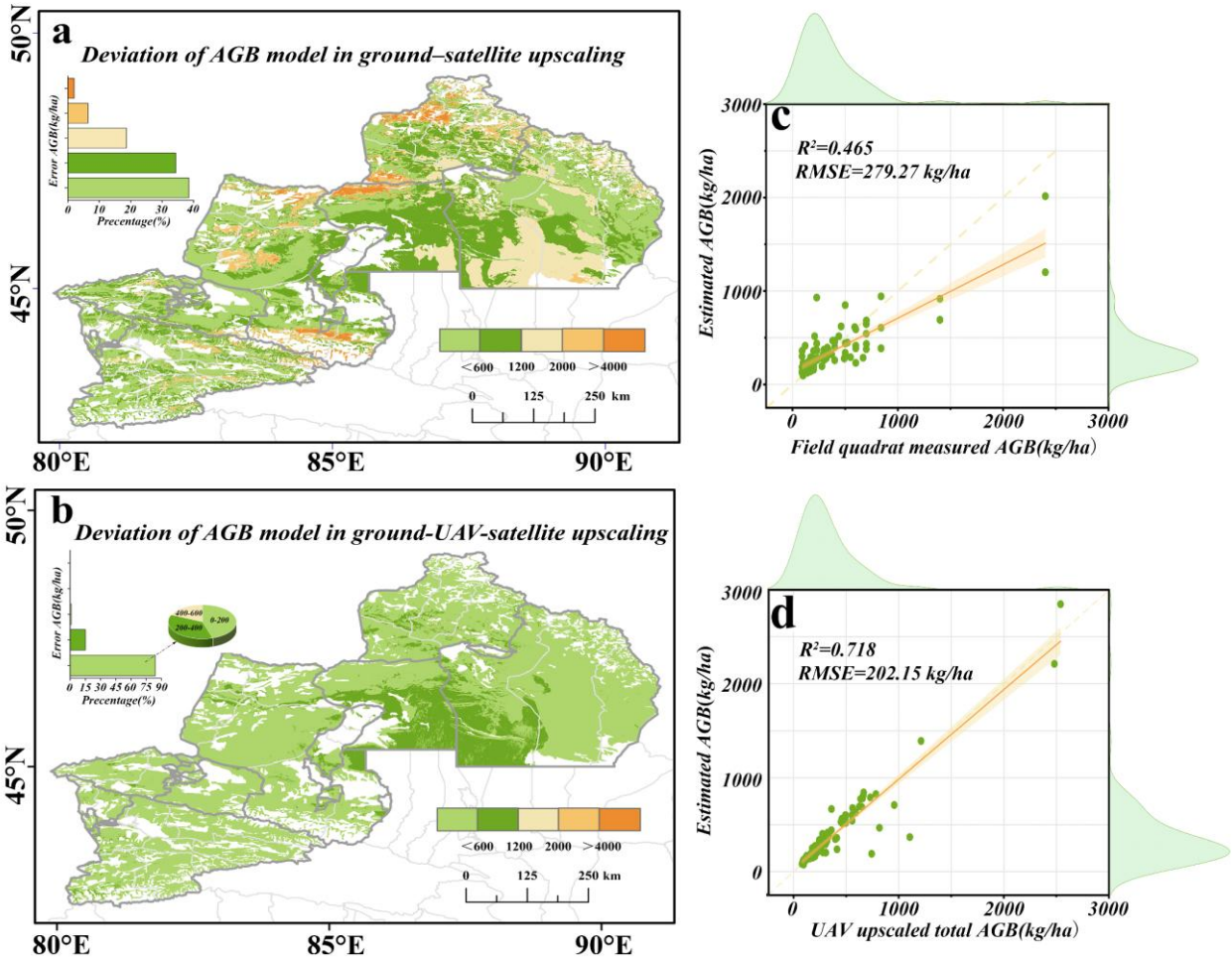


Fig. 11. Comparison of grassland AGB estimation between the ground–satellite method and the ground–UAV–satellite upscaling method. (a, b) Spatial distribution of grassland AGB for the two methods. (c, d) Accuracy comparison of the optimal grassland AGB inversion models.

complexity of vegetation patch boundaries and reflects their spatial structural characteristics [51], while Centroid coordinates describe the spatial locations of vegetation patches and serve as a key indicator for characterizing differences in vegetation spatial patterns [52]. Therefore, the improvement in the accuracy of the post-classification AGB inversion model was not only due to the removal of non-vegetation information, but also benefited from the accurate extraction and inclusion of vegetation geometric and shape features in the modeling process. These features can characterize the spatial structural properties of vegetation and compensate for the limitations of spectral information alone, thereby improving the model's ability to represent biomass differences.

B. Ecological Interpretation of Key Variables for Satellite-Scale Grassland AGB Inversion

In this study, multi-source variables at the satellite scale were systematically screened using the Monte Carlo method, and an optimal ResNet model containing seven key variables was constructed, thereby improving the inversion accuracy of grassland AGB. The screening results showed that geographic factors, including Longitude and Slope, made prominent contributions to the model, reflecting the fundamental control of environmental gradients on the spatial distribution of grassland AGB at the regional scale. As an integrated spatial variable, Longitude can effectively reflect the environmental heterogeneity of northwestern Xinjiang, which is associated with spatial differences in elevation, precipitation, and thermal conditions [53]. Slope constrains grassland AGB accumulation by affecting surface runoff processes, soil moisture redistribution, and nutrient retention efficiency. In general, gentle terrain can reduce soil erosion, promote the conservation of water and nutrients, and provide stable resource conditions for vegetation growth [54]. In contrast, steep slopes may intensify soil and water loss, thereby limiting root development and aboveground biomass accumulation. This result further confirms that topographic conditions remain key limiting factors for AGB in arid grassland ecosystems.

In terms of spectral and soil factors, NLI, as a spectral indicator reflecting vegetation greenness and photosynthetic capacity, can effectively characterize the potential biomass accumulation capacity of vegetation and is an important ecological indicator of vegetation growth status [55]. RTP, as an important variable representing soil nutrient status, contributed to the model, reflecting the long-term regulatory

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effect of soil fertility on grassland productivity. Phosphorus affects vegetation growth rate and biomass accumulation efficiency by participating in plant energy metabolism and material synthesis processes. Its stable contribution also confirms the fundamental supporting role of soil conditions in regional-scale AGB formation [56]. Climatic factors, including pre8-1 and tmp8-5, served as auxiliary driving variables and further revealed the regulatory effects of hydrothermal conditions on grassland AGB in arid areas. Among them, pre8-1 corresponds to the key water-supply stage during grassland regreening and rapid growth, and its contribution indicates that the model effectively captured the role of precipitation in arid areas [57]. tmp8-5 regulates vegetation photosynthetic efficiency and the degree of water stress through its synergistic relationship with precipitation, thereby affecting biomass accumulation efficiency.

The SHAP analysis further revealed the direction of the effects of each variable on AGB prediction. NLI was the most important predictor in the model, and its SHAP values showed a clear nonlinear response with changes in the variable, indicating that the relationship between vegetation spectral characteristics and biomass was jointly influenced by multiple factors, such as canopy structure, vegetation coverage, and background information [58]. The SHAP values of Longitude were mostly negative within the range of 83°–85°, suggesting that this region may experience insufficient water supply or temperatures deviating from the optimal range for vegetation growth, thereby inhibiting vegetation photosynthesis and growth. The negative effect of Slope further verified the adverse influence of steep terrain on water and nutrient accumulation. The SHAP values of RTP showed a gentle increasing trend, indicating that soil phosphorus content had a gradual promoting effect on biomass [59]. The contribution of climatic factors was relatively moderate, suggesting that hydrothermal conditions were not the dominant limiting factors for vegetation growth in the study area. However, their stable positive contribution still supported the basic growth requirements of vegetation, which is consistent with the relatively low contribution of climatic factors in the RF-MDI results.

C. Advantages and Performance of the Ground-UAV-Satellite AGB Inversion Model

This study developed a grassland aboveground biomass estimation framework for northwestern Xinjiang by integrating ground plots, UAV, and satellite remote sensing data. The results showed that, compared with the ground–satellite method, the introduction of UAV-derived samples effectively improved the accuracy and stability of the AGB inversion model [60], [61]. The improvement in model performance mainly resulted from the advantages of UAV data in spatial resolution and spectral information acquisition, which enabled UAV observations to serve as an effective cross-scale bridge between ground observations and satellite remote sensing.

Specifically, in terms of spatial scale, ground plots usually reflect only the measured AGB information within the quadrat extent, whereas satellite pixels cover a larger area and are often jointly affected by vegetation, bare soil, and other background factors. If ground-measured values are directly matched with satellite pixels for model construction, the plot-level information may not adequately represent the actual vegetation conditions within the satellite pixel, thereby introducing errors into model training. UAV images have centimeter-level spatial resolution and can correspond more accurately to ground-measured plots while providing intermediate-scale information between the plot scale and the satellite-pixel scale. By first estimating AGB at the UAV scale and then using the UAV-scale AGB results as modeling samples at the satellite scale, the scale difference between ground plots and satellite pixels can be alleviated to some extent, thereby reducing modeling errors caused by scale inconsistency. In terms of spectral reflectance acquisition, the UAV platform flies at a relatively low altitude and is less affected by atmospheric scattering and complex atmospheric correction processes, enabling it to obtain multispectral reflectance information that is closer to the actual surface conditions. Compared with satellite remote sensing data, UAV imagery has certain advantages in spectral fidelity and ground-object identification capability [62]. For arid grasslands, vegetation coverage is relatively low and the bare-soil background is prominent. Satellite pixels often contain mixed information from vegetation and non-vegetation backgrounds, which can weaken the vegetation spectral signal. UAV images can more clearly capture vegetation canopy spectral characteristics and their spatial variations, thereby providing more reliable input information for AGB estimation. Therefore, using UAV-scale AGB inversion results as training samples for satellite-scale modeling can provide a more suitable and reliable dependent variable for satellite-scale grassland AGB inversion, thereby improving model stability and predictive performance at the regional scale.

Recent related studies have also demonstrated the important role of UAVs in cross-scale remote sensing modeling. Kolluru et al. [63] used sUAS imagery as a bridge between ground plots and Sentinel-2 satellite data for cross-scale mapping of grassland canopy cover and AGB. Xiong et al. [64] also showed that UAV–satellite synergy can alleviate the problems of limited ground samples and scale mismatch by generating upscaled samples, thereby improving regional-scale prediction performance. Compared with these studies, the present study further applied the bridging role of UAVs to AGB estimation in arid grasslands. Arid grasslands are characterized by sparse vegetation, obvious spatial patchiness, and strong interference from bare-soil backgrounds on spectral signals. Direct matching between ground plots and satellite pixels is therefore susceptible to scale differences and background interference. To address this issue, this study adopted a stepwise ground–UAV–satellite modeling strategy. UAV multispectral images were first used to estimate UAV-scale AGB, and the resulting UAV-scale AGB estimates were then used as important training samples for satellite-scale modeling. In this way, ground-measured information was transitioned through the intermediate UAV scale before being extended to the satellite scale, thereby enhancing the representativeness of training samples and the stability of regional AGB inversion.

In terms of model algorithms, this study compared two ML algorithms, RF and XGBoost, and one DL algorithm, ResNet.

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The results showed that ResNet achieved better predictive performance at both the UAV and satellite scales. This may be attributed to the residual connection structure used in ResNet, which enables the model to effectively retain original feature information while increasing network depth and alleviates the gradient vanishing problem, thereby allowing the model to more fully learn the nonlinear relationships between multi-source remote sensing variables and AGB [65]. Compared with RF and XGBoost, which are based on decision-tree ensembles, ResNet showed stronger feature representation capability and model stability when handling high-dimensional, multi-scale, and strongly correlated feature combinations. This indicates its potential advantage in grassland AGB inversion involving multiple remote sensing variables. Overall, this study further demonstrates the bridging role of UAVs in cross-scale AGB estimation for arid grasslands. By introducing UAV-scale inversion results and using the ResNet deep learning model, a more continuous scale transformation relationship was established between ground plots and satellite pixels, improving the stability and reliability of model prediction. This framework provides an effective approach for regional monitoring of grassland AGB in arid areas and offers methodological support for subsequent grassland productivity assessment and ecosystem carbon cycle research.

D. Limitations and Future Work

This study constructed a ground–UAV–satellite-based grassland AGB inversion model and achieved relatively stable estimation results; however, several limitations remain. Although variable selection and multi-model comparison can reduce the risk of overfitting in small-sample modeling, the current sample size is still relatively limited for deep learning models. Future studies will continue to increase the number of ground quadrats and UAV observations to further improve model applicability and stability. In terms of spatial-scale matching, this study established 16 UAV imaging positions within each 250 m × 250 m plot and used the mean band reflectance and corresponding vegetation indices from the 16 imaging positions, each covering approximately 26 m × 35 m, to approximate UAV spectral features at the plot scale. Similar approaches have been applied in grassland remote sensing studies, such as setting 16 UAV subplots within a 250 m × 250 m monitoring plot for MODIS pixel-scale validation and using the FragMAP system to match the 250 m × 250 m UAV observation scale with MODIS 250 m pixels [66]. However, this approach does not represent full UAV coverage of the entire 250 m × 250 m plot. Strictly speaking, it remains an approximate representation and may introduce some uncertainty because the UAV observation extent is not completely consistent with the satellite pixel coverage. Future studies will acquire UAV orthomosaic images covering the full satellite pixel extent to further reduce the uncertainty caused by incomplete scale matching.

In terms of data acquisition, the current UAV data mainly consist of multispectral images. In future work, multiple UAV sensors, such as RGB, hyperspectral, and thermal infrared sensors, can be jointly used to obtain richer spectral and texture information, thereby improving the accuracy and robustness of AGB estimation [67]. From the temporal perspective, the field surveys in this study were mainly conducted during the peak growing season of grasslands, while changes in vegetation structure and biomass characteristics during the senescence period and other growth stages were not considered. This limits the ability of the model to characterize the full annual growth process of grasslands. Therefore, continuous multi-period and multi-year observations should be conducted in future studies to establish a more comprehensive time-series monitoring system. In addition, although the YOLOv8n-Seg model performed well in vegetation classification, its recognition accuracy in sparsely vegetated areas and mixed-pixel regions still needs further improvement. More refined semantic segmentation algorithms, such as UNet++ and DeepLabV3+, can be introduced in future studies to optimize the vegetation extraction process and provide a more accurate classification basis for AGB inversion [68], [69]. Overall, future research can establish a more accurate and robust remote sensing monitoring system for grassland AGB by increasing the number of ground quadrats, integrating multi-sensor UAV data, constructing temporal dynamic models across multiple years and different growth stages, and optimizing vegetation classification methods. This will provide more reliable technical support for grassland ecosystem management.

VI. Conclusions

This study proposed a ground–UAV–satellite cross-scale collaborative estimation method. Using high-resolution UAV observations as an intermediate scale between ground plots and satellite pixels, this method integrated vegetation classification, variable selection, and multi-model comparison to construct a multi-scale AGB estimation system for arid grasslands during the peak growing season. The main conclusions are as follows:

1) In terms of vegetation classification from UAV images, YOLOv8n-Seg outperformed RF and CART across the three grassland types, namely steppe, meadow, and desert, and could more accurately extract vegetation areas in arid grasslands.
2) In terms of variable selection and model construction, the models based on selected factors achieved better performance with a mean R^2 of 0.662 than the models constructed using all 79 driving variables, which had a mean R^2 of 0.637. This indicates that Monte Carlo screening can reduce the influence of redundant variables on model performance.
3) In terms of cross-scale estimation, the UAV-scale AGB inversion results were used as satellite-scale modeling samples to construct the ground–UAV–satellite cross-scale estimation model. This model achieved higher accuracy (R^2 = 0.718, RMSE = 202.15 kg/ha) than the ground–satellite-scale estimation model (R^2 = 0.465, RMSE = 279.27 kg/ha), providing a feasible approach for cross-scale monitoring of grassland AGB in arid areas during the peak growing season.

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